

Did Malta's PM2.5 death rate fall by two-thirds?

The EEA's own estimate, tested against its data and Malta's stations

THE CLAIM

“... is estimated to have been reduced by 67.7% between 2005 and 2023 ...”

European Environment Agency, Air pollution quick country facts (country fact sheets 2025), 1 Dec 2025

Malta, deaths attributable to PM2.5 above 5 µg/m³ per 100 000 aged 30+: 143.5 → 46.3; 172 deaths in 2023.

A modelled estimate: a WHO risk function applied to a concentration map and death statistics.

VERDICT

LARGELY SUPPORTED

Reported exactly; the 2005 start is a map value.

SUPPORTED

LARGELY
SUPPORTED

NOT
SUBSTANTIATED

MISLEADING

CONTRADICTED

MAIN RESULTS

67.7%

The figures match

143.5 → 46.3 per 100 000 aged 30+, in the EEA's own table; 172 deaths in 2023, as Eurostat.

≤1.2%

The method reproduces

From Eurostat's death counts we get 346 and 170 deaths, against the EEA's 348 and 172.

–39%

Measured PM2.5 fell

At Msida, 22.7 → 13.9 µg/m³ from 2007 to 2023; three long-running stations –25% since 2010.

0

None reported for 2005

No Maltese PM2.5 reached the EEA for 2005 (first: August 2006); that year's value is a map value.

–51%

Deaths fell by half

348 → 172: the rate fell more because the over-30 population grew 54%.

FAIR TO SAY

The EEA says “is estimated” and names the threshold and age group. Counting all concentrations, or starting in 2007, the fall is about 55–57%; the 95% CI covers only the risk function.

WHAT WOULD SETTLE IT

☐ An uncertainty range for Malta's 2005 exposure.

☐ Why the technical report gives 173, not 172.

☐ Any 2005 PM2.5 data not sent to the EEA.